
AI-Powered Customer Support Automation System using LangGraph

IBM Agentic AI Certification Assignment

Execution Screenshots

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Screenshot 1 – Sales Agent Routing & RAG Retrieval

Description

This screenshot demonstrates the successful classification of a Sales-related query. The LangGraph workflow routes the request to the Sales Agent, retrieves pricing information from the local knowledge base (RAG), and generates the final response using Ollama (Llama 3.2).

```
• (venv) tejas@pop-os:~/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph$ /home/tejas/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph/app.py
Enter Customer Query: What are the pricing plans available for your software?
=====
Customer Query: What are the pricing plans available for your software?
Detected Intent: Sales
Route: Sales Agent

Final Response:

We have three pricing plans available:

1. Starter Plan: $20/month
2. Professional Plan: $50/month

For our Enterprise Plan, please contact our sales team to discuss custom pricing and options.
=====
```

Screenshot 2 – Account Agent Response

Description

This screenshot demonstrates the routing of an Account-related query to the Account Agent, which generates an appropriate response using the local knowledge base.

```
• (venv) tejas@pop-os:~/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph$ /home/tejas/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph/venv/bin/python /home/tejas/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph/app.py
Enter Customer Query: I forgot my account password.
=====
Customer Query: I forgot my account password.
Detected Intent: Account
Route: Account Agent

Final Response:

I'd be happy to help you with resetting your account password.

To do this, I'll need to guide you through a few steps.

Firstly, please check if you have access to the Account Settings page. If you can't find it, don't worry! You can try searching for "Account Settings" in our system or contact us directly.

Once you're in the Account Settings page, look for the option to reset your password. This is usually located under a section called "Security" or "Password Management".

If you need help finding this option, feel free to ask and I'll do my best to assist you.

Also, just a heads up: if you have any trouble resetting your password, our system will send you a password reset email immediately. This email will contain instructions on how to reset your password.
=====
```

Screenshot 3 – Technical Support Agent

Description

This screenshot demonstrates the routing of a Technical Support query. The Technical Agent retrieves relevant troubleshooting information and generates the final response.

```
(venv) tejasmpop-os:~/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph$ /home/tejasmpop/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph/venv/bin/python /home/2/customer-support-langgraph/app.py
Enter Customer Query: My application crashes whenever I upload a file.
=====
Customer Query: My application crashes whenever I upload a file.
Detected Intent: Technical
Route: Technical Support Agent

Final Response:

I'd be happy to help you with that issue.

First, let's try something simple. Have you tried restarting the application? Sometimes, a simple restart can resolve issues like this.

If the problem persists, we'll need to investigate further. Can you please check your internet connectivity while uploading a file? Is it stable and working properly?

If you're still experiencing issues after checking your internet connection, we may need to dig deeper into the application's behavior.
=====
```

Screenshot 4 – Billing Agent with Human-in-the-Loop Approval

Description

This screenshot demonstrates the Human-in-the-Loop workflow. The Billing Agent identifies the refund request as a high-risk operation and routes it to the Supervisor Agent before returning the final approved response.

```
(venv) tejasmpop-os:~/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph$ /home/tejasmpop/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph/venv/bin/python /home/tejasmpop/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph/app.py
Enter Customer Query: I need a refund for my annual subscription.
=====
Customer Query: I need a refund for my annual subscription.
Detected Intent: Billing
Route: Billing Agent → Supervisor Agent

Final Response:

Human Supervisor Approved.

We appreciate your patience as we work to assist you with your refund request. Please provide us with more details regarding the issue, including any relevant order numbers, dates of purchase, or other supporting documentation. We'll do our best to expedite the process once we receive this information.
=====
```

Screenshot 5 – Memory Storage

Description

This screenshot demonstrates that the customer interaction is successfully stored in the SQLite database for future retrieval.

```
• (venv) tejas@pop-os:~/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph$ /home/tejas/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph/venv/bin/python /home/tejas/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph/app.py
Enter Customer Query: I have a billing issue.
=====
Customer Query: I have a billing issue.
Detected Intent: Billing
Route: Billing Agent

Final Response:
I'd be happy to help with your billing issue. Can you please provide more details about the issue you're experiencing, such as what specifically is not being resolved or if there's an error on your bill? This will allow me to better assist you and potentially escalate it to a supervisor for further review and approval if needed.
=====
```

Screenshot 6 – Memory Recall

Description

This screenshot demonstrates the Memory Agent retrieving the customer's previous interaction from the SQLite database and returning it as part of the response.

```
• (venv) tejas@pop-os:~/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph$ /home/tejas/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph/venv/bin/python /home/tejas/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph/app.py
Enter Customer Query: What was my previous support issue?
=====
Customer Query: What was my previous support issue?
Detected Intent: Memory
Route: Memory Agent

Final Response:
Your previous issue was:
I have a billing issue.
=====
```

Screenshot 7 – SQLite Memory Database

Description

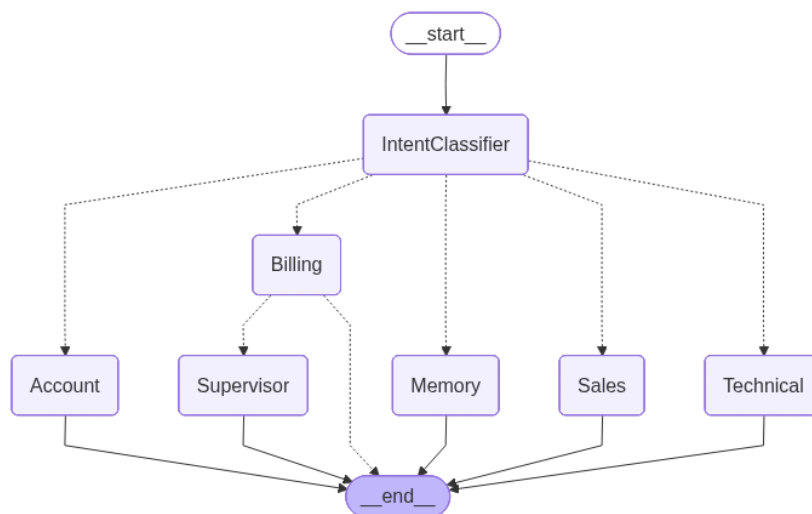
This screenshot verifies the SQLite database used for persistent conversation memory. It shows the stored customer interactions that are used by the Memory Agent.

```
(venv) tejas@pop-os:~/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph$ sqlite3 memory.db
SQLite version 3.37.2 2022-01-06 13:25:41
Enter ".help" for usage hints.
sqlite> .tables
conversations
sqlite> SELECT * FROM conversations;
1|I have a billing issue.
2|What was my previous support issue?
3|I need a refund.
4|I need my invoice.
5|I need a refund.
6|i need a refund
7|What are the pricing plans available for your software?
8|What are the pricing plans available for your software?
9|What are the pricing plans available for your software?
10|I forgot my account password.
11|My application crashes whenever I upload a file.
12|I need a refund for my annual subscription.
13|I have a billing issue.
14|What was my previous support issue?
sqlite> █
```

Screenshot 8 – LangGraph Workflow Diagram

Description

This workflow diagram illustrates the complete LangGraph architecture, including intent classification, conditional routing, specialized support agents, Human-in-the-Loop approval, Supervisor Agent, and workflow termination.



Screenshot 9 – Project Structure

Description

This screenshot shows the final project structure, including all source files, knowledge base documents, workflow diagram, SQLite database, and supporting resources submitted as part of the assignment.

```
(venv) tejasm@pop-os:~/VIT/7th_sem/ibm_course/ass2/customer-support-langgraph$  
tree -L 2 -I "venv|__pycache__"  
Customer Support LangGraph Project Structure  
.  
├── agents.py  
├── app.py  
├── data  
│   ├── faq.txt  
│   ├── policy.txt  
│   ├── pricing.txt  
│   └── technical.txt  
├── graph.py  
├── llm.py  
├── memory.db  
├── memory.py  
├── rag.py  
├── README.md  
├── requirements.txt  
├── screenshots  
│   ├── account_agent.png  
│   ├── billing_human_approval.png  
│   ├── memory_recall.png  
│   ├── memory_storage.png  
│   ├── project_structure.png  
│   ├── sales_agent.png  
│   ├── sqlite_database.png  
│   ├── technical_support.png  
│   └── workflow.png  
├── state.py  
├── supervisor.py  
└── workflow.py  
  
2 directories, 25 files
```